Pranav Anil Ingle

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.37
Shri Dawale Junior College, Akola	Class XII (Maharashtra State Board)	2022	79.67%
Prabhat Kids School, Akola	Class X (CBSE)	2020	90.60%

SKILLS

Programming Languages: C++, C, Python, Java, SQL (MySQL)

Cloud & Infrastructure: AWS (EKS, DynamoDB, IAM, SigV4), Microsoft Azure, Kubernetes (RBAC), Docker

Backend & Distributed Systems: Distributed Systems, REST APIs, Microservices Architecture, Event-Driven Architecture, Webhooks, Multi-threading, Socket Programming (TCP/IP), Consistent Hashing

System Security & Protocols: OAuth2, AWS SigV4, Kubernetes RBAC, Model Context Protocol (MCP), Least-Privilege IAM, Response Sanitization

Data & Storage: DynamoDB, Relational Databases, Database Transaction Management, Data Modeling (Hierarchical & Key-Value), Write-Ahead Logging (WAL), Object Pool Allocation

AI & Integration: Claude AI, Enterprise LLM Integration, Microsoft Graph API

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib

Developer Tools & Systems: Linux System Calls (epoll, mmap), Linux/Unix Shell, Git

WORK EXPERIENCE

Software Development Engineering Intern | Kotak Mahindra Bank

- Engineered scalable **cloud infrastructure automation**, internal developer tooling, and enterprise AI integrations across AWS, Azure, and Kubernetes to eliminate manual operational overhead, applying **secure systems design principles** such as OAuth2, SigV4, IAM, and RBAC across multi-tenant services.
- Re-architected an open-source AWS Model Context Protocol (MCP) proxy into a **centralized, stateless service** deployed on Amazon EKS with HTTP communication, enabling engineers to query cluster diagnostics, pod states, and logs using natural language via Claude AI; built an **automated response-sanitization layer** to redact sensitive tokens/URLs and implemented dynamic SigV4 credential refresh to ensure uninterrupted sessions without exposing credentials.
- Designed a centralized IP management backend on Amazon DynamoDB using integer-based IP indexing and hierarchical parent-child data models for **low-latency CIDR overlap detection**; developed 5 production REST APIs to automate cell creation, sequential subnet allocation, and legacy tenant onboarding, successfully eliminating manual spreadsheet tracking and network provisioning conflicts.
- Built an asynchronous, event-driven email service using Microsoft Graph API webhooks and Azure app registrations to capture customer queries in near-real time; integrated internal LLM APIs (Kompanion AI) over OAuth2 with automatic token refresh, structuring a hybrid routing engine that **autonomously answers routine queries** while **drafting responses for complex escalations** requiring human review.

PROJECTS

High-Performance Trading Order Book

- Architected a deterministic, ultra-low-latency limit order book matching engine in C++ capable of processing market data streams with sub-microsecond execution time.
- Eliminated runtime heap-fragmentation and OS kernel overhead in the critical path by implementing a custom object pool allocator, completely bypassing dynamic new/malloc calls.
- Optimized data structures using an unrolled linked list combined with a custom hash map to achieve stable O(1) time complexity for order insertions, cancellations, and price-level lookups.

- Enforced cache-line alignment and structural bit-packing across core message structs to maximize CPU L1/L2 cache hits during rapid price matching.

Fault-Tolerant Distributed Key-Value Store

- Designed and implemented a high-availability, distributed in-memory key-value store utilizing a custom Consistent Hashing Ring to distribute keys evenly across dynamic server nodes.
- Engineered an automated masterless failover mechanism featuring heartbeat detection and dynamic data replication, ensuring zero data loss and continuous uptime during single-node crashes.
- Developed a lightweight, binary wire protocol over TCP sockets for inter-node communication, reducing network serialization payload overhead compared to traditional HTTP/REST architectures.
- Implemented a sequential Write-Ahead Log (WAL) using memory-mapped files (mmap) to ensure strict data persistence and rapid state recovery upon node reboots.

High-Throughput Inventory & Transaction Engine

- Designed and built a highly concurrent inventory transactional system handling simulated concurrent purchase requests using multi-threaded execution pools.
- Optimized database interactions by establishing a MySQL connection pool, reducing connection latency overhead during peak query volumes.
- Resolved inventory race conditions and double-spending vulnerabilities by implementing strict pessimistic locking mechanisms and tuning transactional isolation levels.
- Engineered an asynchronous analytics engine that processed historical order data using indexed aggregate SQL queries to track running performance metrics under heavy indexing.